

Useful Memory Has a Horizon

Temporal Validity, Roughness, and Structural Limits for
Finite-Memory Tracking under Drift

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Abstract

Memory is not always an asset under drift. Past data reduce sampling noise, but as the target distribution moves, retaining more evidence also accumulates temporal misalignment with the present. Finite-memory distribution tracking is therefore governed by a temporal-validity problem: how long can retained evidence improve estimation of the present before drift-induced misalignment overtakes variance reduction?

We answer this question by separating finite-memory error into the carrier of the chosen geometry and the roughness-controlled staleness of the drift path. This yields a carrier-roughness useful-memory horizon law,

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H,$$

where a is the carrier exponent and $H \in (0, 1]$ is the roughness exponent. Optimizing this balance gives the useful-memory scale $n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}$ and the corresponding optimized error law.

We instantiate the law with a rigorous minimum kernel: a one-dimensional bounded-support fixed-span triangular array in which a quantile/Bahadur argument gives the canonical $a = 1/2$ carrier. A Gaussian witness lower bound shows that the resulting horizon is structural in this canonical regime, and a refined witness analysis gives the exact Gaussian testing frontier and identifies an endpoint-minimal Hölder witness that improves the ramp constant for $H < 1$. The same decomposition also identifies regimes where useful memory persists beyond the minimum kernel: low intrinsic dimension can preserve the raw W_2 carrier, while fixed- ε Sinkhorn geometry provides an operational carrier with stable finite-sample behavior.

The resulting viewpoint is adjacent to dynamic regret and adaptive windowing, but the object is different: temporal validity of retained evidence. Useful memory has a path-dependent horizon, and retained evidence can become invalid for the present before changepoint evidence becomes statistically visible.

Keywords: tracking under drift; useful memory; temporal validity; carrier-roughness useful-memory horizon law; invalidity gap; Wasserstein distance.

MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

In stationary statistics, memory is an unambiguous resource: larger samples reduce variance because the target distribution does not change. Under drift, that logic fails. In streaming estimation, monitoring, and online adaptation, the evidence retained in memory was generated by distributions that may no longer represent the present. More memory therefore does two things at once: it reduces finite-sample noise, and it increases the age of the evidence used to estimate the current state.

That tension creates a horizon problem. Short windows are fresh but noisy. Long windows are stable but stale. The relevant question is not only whether memory helps, but how long retained evidence can remain temporally valid before drift-induced misalignment overtakes the remaining variance gain. The paper studies that question in distributional geometry and treats the horizon of useful memory as the main statistical object.

The central contribution is a carrier-roughness useful-memory horizon law. The starting point is an abstract decomposition

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H,$$

where $a > 0$ is the finite-sample carrier exponent of the chosen geometry and sample regime, $H \in (0, 1]$ is the temporal roughness exponent of the drift path, and ζ is the roughness amplitude. Optimizing this balance yields the useful-memory scale

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The cube-root law is recovered as the special case $a = 1/2$, $H = 1$, corresponding to a root- n carrier and worst-case linear staleness. The paper is therefore not about the cube-root law alone. It is about the larger law that explains when cube-root scaling appears, when it does not, and why the horizon is path dependent.

To make this law concrete, the paper proceeds in three steps.

First, it identifies a *minimum kernel*: a one-dimensional bounded-support fixed-span triangular-array setting in which the carrier can be instantiated rigorously. In that regime, the quantile representation of W_2^2 , a triangular-array Bahadur expansion, and a design-effect comparison yield the canonical carrier

$$\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2}).$$

This supplies a concrete $a = 1/2$ regime of the abstract law.

Second, the paper identifies an *extended regime*: low-dimensional or low-intrinsic-dimensional settings in which the triangular-array carrier tracks the i.i.d. mixture benchmark with only a constant-level gap. This extends the theory beyond the one-dimensional proof regime without pretending to solve the full high-dimensional raw- W_2 problem.

Third, it studies an *operational regime*: fixed- ε Sinkhorn geometry as an operational carrier whose finite-sample behavior appears stable even where raw Wasserstein geometry is dimensionally fragile. This regime is included as an operational extension of the same horizon law, not as a claim of full theorem-level generality.

The theory is complemented by a structural lower bound. On the canonical carrier regime $a = 1/2$, a Gaussian witness construction shows that the useful-memory scale is not merely the optimum of one procedure. The refined witness

analysis also identifies the exact Gaussian testing frontier and shows that, for $H < 1$, the usual ramp witness is not the lowest-energy endpoint-saturating Hölder shape. Thus the same lower-bound mechanism supplies both the structural exponent law and a sharper picture of the constants inside the canonical regime.

The empirical logic mirrors the theory. The paper first makes the basic variance–staleness trade-off visible through a finite-memory U-curve. It then shows roughness-dependent horizon scaling across path classes, carrier behavior beyond the minimum kernel, and a measurable invalidity gap between temporal validity and changepoint evidence. The central conceptual distinction is simple: temporal validity is not changepoint evidence. A detector can remain statistically silent while retained evidence is already stale for the present.

This framing is adjacent to several literatures—dynamic regret, adaptive windowing, locally stationary smoothing, adaptive filtering, and distributional tracking under transport geometry—but the paper asks a different question. Dynamic-regret analysis asks how well one tracks a moving comparator, and adaptive-window or drift-detection methods ask when the accumulated evidence is sufficient to react. The question here is different: for finite-memory tracking under drift, how long does retained evidence remain temporally valid for estimating the present distribution before age becomes the dominant source of error?

The paper proceeds as follows. Section 2 develops the carrier-roughness useful-memory horizon law, including the minimum-kernel result, the structural lower bound, the refined Gaussian witness geometry, and the broader extended and operational regimes. Section 3 presents the corresponding finite-sample evidence. Section 4 discusses related work, and Section 5 discusses scope, limitations, and open problems.

2 The Carrier-Roughness Useful-Memory Horizon Law

Under drift, memory is both a statistical resource and a temporal liability. Retaining more data reduces finite-sample noise, but it also increases the age of the evidence used to represent the present. The object of interest is therefore not a detector threshold or a particular adaptive rule, but the error of finite-memory representation itself.

For a finite-memory estimator with lag weights $w_0, \dots, w_{n-1}$, the natural age variable is the effective lag

$$A(w) := \sum_{j=0}^{n-1} j w_j.$$

Its corresponding staleness at time t is the weighted transport misalignment

$$S_t(w) := \sum_{j=0}^{n-1} w_j W_2(P_{t-j}, P_t).$$

For a uniform window, $A(w) = (n-1)/2$, so window length and temporal age are the same object up to constants. This is why memory length becomes a statistical question rather than a purely algorithmic one.

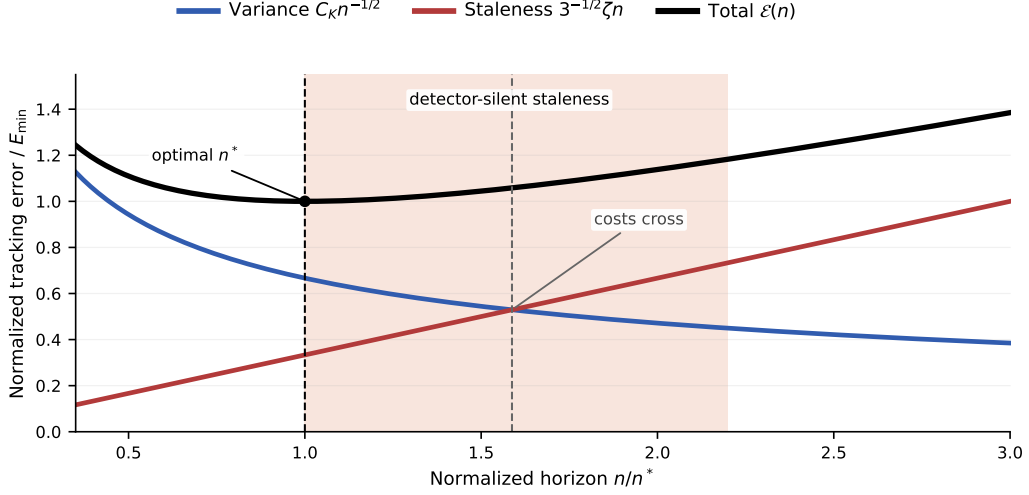


Figure 1: Two clocks of drift. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the useful-memory horizon n^* . The shaded region represents detector-silent staleness: retained evidence can lose temporal validity before enough changepoint evidence accumulates to trigger an alarm.

2.1 Abstract Upper Law

The abstract theorem separates finite-sample carrier behavior from drift-induced staleness.

Theorem 2.1 (Abstract upper law). *Let d be a probability metric satisfying the triangle inequality. At time t , let P_t be the present target law, let $\bar{P}_t^{(n)}$ be the window target, and let $\hat{P}_t^{(n)}$ be the estimator built from the last n observations. Assume that for some constants $C_K > 0$, $C_S > 0$, exponent $a > 0$, roughness index $H \in (0, 1]$, and amplitude $\zeta > 0$,*

$$\mathbb{E} d\left(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) \leq C_K n^{-a}$$

and

$$d\left(\bar{P}_t^{(n)}, P_t\right) \leq C_S \zeta n^H$$

hold for all n in the regime of interest. Then

$$\mathbb{E} d\left(\hat{P}_t^{(n)}, P_t\right) \leq C_K n^{-a} + C_S \zeta n^H.$$

Proof. By the triangle inequality,

$$d\left(\hat{P}_t^{(n)}, P_t\right) \leq d\left(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) + d\left(\bar{P}_t^{(n)}, P_t\right).$$

Taking expectations and applying the two assumptions yields the claim. $\square$

The proof is immediate, but the theorem makes the modular structure explicit: the path class supplies the staleness term, while the chosen geometry and sample regime supply the carrier term.

Corollary 2.2 (Optimized useful-memory horizon law). *Under the assumptions of theorem 2.1, the upper envelope*

$$\Phi(n) := C_K n^{-a} + C_S \zeta n^H$$

is minimized, up to constant factors, at

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)},$$

and the corresponding optimized error scale is

$$\Phi(n^*) \asymp C_K^{H/(a+H)} \zeta^{a/(a+H)}.$$

Proof. Balance the decreasing carrier term against the increasing staleness term:

$$C_K n^{-a} \asymp C_S \zeta n^H.$$

This gives $n^{a+H} \asymp C_K/\zeta$, hence the displayed horizon law. Substituting that scale back into either term gives the optimized error scale. $\square$

The cube-root law is recovered when $a = 1/2$ and $H = 1$. More generally, the carrier exponent and the roughness exponent jointly determine the useful-memory scale. There is no universal horizon independent of those two ingredients.

2.2 Path Classes and the Roughness Index

The staleness side of the law is controlled by a deterministic path class. For $H \in (0, 1]$ and $\zeta > 0$, write

$$\mathfrak{P}_H(\zeta) := \{(P_s)_{s \in \mathbb{Z}} : W_2(P_{t-j}, P_t) \leq \zeta j^H \text{ for all } j \geq 0\}.$$

The case $H = 1$ is the local Lipschitz envelope; smaller H correspond to slower accumulation of staleness with lag.

Lemma 2.3 (Hölder staleness bound via averaged couplings). *Let*

$$\bar{P}_t^{(n)} := \frac{1}{n} \sum_{j=0}^{n-1} P_{t-j}.$$

If the target path belongs to $\mathfrak{P}_H(\zeta)$, then

$$W_2^2(\bar{P}_t^{(n)}, P_t) \leq \frac{\zeta^2}{n} \sum_{j=0}^{n-1} j^{2H},$$

and therefore

$$W_2(\bar{P}_t^{(n)}, P_t) \leq C_{H,n} \zeta n^H,$$

where

$$C_{H,n} := \left(n^{-(2H+1)} \sum_{j=0}^{n-1} j^{2H} \right)^{1/2}.$$

In particular, $C_{H,n} \rightarrow (2H+1)^{-1/2}$ as $n \rightarrow \infty$.

Proof. For each lag j , let π_j be an optimal coupling between P_{t-j} and P_t . Their average

$$\bar{\pi} := \frac{1}{n} \sum_{j=0}^{n-1} \pi_j$$

couples $\bar{P}_t^{(n)}$ with P_t , so

$$W_2^2(\bar{P}_t^{(n)}, P_t) \leq \int \|x - y\|^2 d\bar{\pi}(x, y) = \frac{1}{n} \sum_{j=0}^{n-1} W_2^2(P_{t-j}, P_t).$$

Applying the Hölder envelope gives the displayed finite- n constant, and taking square roots completes the proof. $\square$

Combining lemma 2.3 with corollary 2.2 yields the family

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

On the canonical statistical regime $a = 1/2$,

$$n^*(1/2, H) \asymp (C_K/\zeta)^{2/(1+2H)}.$$

At $H = 1$, this recovers the cube-root horizon; at $H = 1/2$, it yields the intermediate $n^* \asymp C_K/\zeta$ regime. These are members of the same law, not separate theorem stories. The finite- n constant in lemma 2.3 is the constant used throughout the paper for uniform windows; the simpler $(2H + 1)^{-1/2}$ expression is its large-window limit.

2.3 Minimum Kernel: A Rigorous Carrier Instantiation

The abstract law needs at least one rigorous carrier instantiation. The minimum kernel is the bounded-support fixed-span one-dimensional triangular array. It provides the first rigorous carrier entering the abstract horizon law.

Proposition 2.4 (Minimum-kernel carrier). *Let $X_{n,1}, \dots, X_{n,n}$ be independent one-dimensional observations with laws $P_{n,1}, \dots, P_{n,n}$, and define the window mixture*

$$\bar{P}_n = \frac{1}{n} \sum_{j=1}^n P_{n,j}.$$

Assume:

- (i) *all $P_{n,j}$ are supported on a common compact interval $[L, U]$;*
- (ii) *the within-window drift span is uniformly bounded in n ;*
- (iii) *on an interior quantile band $u \in [\varepsilon, 1 - \varepsilon]$, the mixture density $\bar{f}_n$ is bounded below and uniformly Hölder;*
- (iv) *the triangular-array empirical quantile process admits a Bahadur representation with integrated remainder $o(n^{-1})$ on that interior band.*

Then

$$\mathbb{E} W_2^2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1}),$$

and therefore

$$\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2}).$$

Proof sketch. In one dimension,

$$W_2^2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = \int_0^1 (\hat{q}_n(u) - q_n(u))^2 du,$$

so the problem reduces to the integrated L_2 error of the empirical quantile process. On the interior band, the Bahadur representation gives

$$\hat{q}_n(u) - q_n(u) = \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u).$$

The leading term has integrated variance $O(n^{-1})$ because the observations are independent and the fixed-span triangular design changes only constants relative to the i.i.d. mixture benchmark. The remainder splits into an empirical increment term and a Taylor term; the empirical increment term is the dominant one, but under the stated interior regularity it still satisfies an integrated $o(n^{-1})$ bound. Bounded support makes the boundary-band contribution lower-order. Summing the leading term, the lower-order cross term, and the integrated remainder gives $\mathbb{E}W_2^2 = O(n^{-1})$, and Jensen yields the displayed W_2 rate.

The main conclusion is the exponent. Under this bounded-support fixed-span regime, the triangular-array carrier remains root- n and differs from the i.i.d. mixture benchmark only through a design effect.

2.4 A Structural Lower Bound

The upper law would be substantially weaker if it were only the optimum of one estimator. The lower bound shows that, on the canonical carrier regime $a = 1/2$, the useful-memory scale is imposed by the drift geometry itself.

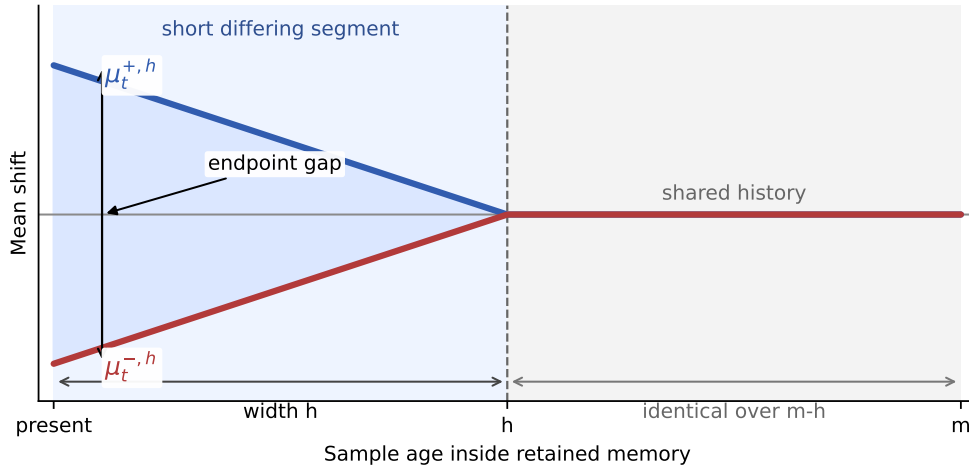


Figure 2: Gaussian witness for the structural lower bound. The two paths share most of the retained history and differ only on a recent segment of width h , yet remain separated at the present. Optimizing that recent segment already produces the same horizon exponent as the upper law on the canonical carrier regime.

Proposition 2.5 (Structural lower bound on the canonical carrier regime). *Fix $H \in (0, 1]$ and work on the Gaussian location subclass with known variance σ^2 and witness paths*

$$\mu_{t-j}^{\pm, h} = \pm \beta (h - j)_+^H, \quad j = 0, 1, \dots, h,$$

where $\beta \leq \zeta$ enforces the roughness budget. Then there exists a constant $c_H > 0$ depending only on H such that the endpoint tracking risk over this subclass satisfies

$$\inf_{\hat{P}} \sup_{P \in \mathcal{G}_{H, \zeta}} \mathbb{E} W_2(\hat{P}_t, P_t) \geq c_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Consequently, on the canonical carrier regime $a = 1/2$, the useful-memory scale

$$h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$$

is structural even on this restricted subclass.

Proof sketch. Compare two hypotheses generated by the witness paths $\mu_{t-j}^{+, h}$ and $\mu_{t-j}^{-, h}$. At the present time, the endpoint separation is of order βh^H . On the Gaussian location subclass, this is also the W_2 separation. The Kullback–Leibler divergence between the two hypotheses is proportional to

$$\beta^2 \sum_{r=1}^h r^{2H} / \sigma^2.$$

Choosing

$$\beta = \min \left\{ \zeta, \frac{\sigma}{2 \sqrt{\sum_{r=1}^h r^{2H}}} \right\}$$

keeps the testing problem in the Le Cam regime while respecting the roughness budget. The resulting two-point lower bound is of order βh^H . Balancing the active roughness constraint against the testing constraint gives $h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$ and therefore the displayed lower law.

At $H = 1$, this recovers the cube-root horizon and the lower scale $\sigma^{2/3} \zeta^{1/3}$. The lower bound is subclass based rather than class-tight, but that is enough for the central claim: the horizon is not just an estimator artifact.

The proposition above is the lower-bound fact needed for the main horizon law. The same witness analysis also exposes a sharper constant-level geometry. We record those refinements next because they explain what is intrinsic to the canonical $a = 1/2$ regime and what was only an artifact of the original ramp proof with its testing relaxation.

2.5 Refined Geometry of the Gaussian Witness

Let Φ and φ denote the standard normal distribution function and density. For the ramp witness in proposition 2.5, exact Gaussian testing gives a one-dimensional variational problem rather than a Pinsker-style constant.

Proposition 2.6 (Exact Gaussian ramp frontier). *Fix $H \in (0, 1]$ and set*

$$p_H := \frac{2H}{2H+1}.$$

Let $x_H > 0$ be the unique maximizer of $x^{p_H}\Phi(-x)$, equivalently the solution of

$$p_H\Phi(-x_H) = x_H\varphi(x_H).$$

For the Gaussian ramp witness

$$\mu_{t-j}^{\pm, h} = \pm\beta(h-j)_+^H,$$

the exact two-point lower bound has asymptotic constant

$$C_H^{\text{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H),$$

with horizon shape

$$h_H^* \sim (\sqrt{2H+1} x_H)^{2/(2H+1)} \left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)}.$$

Proof sketch. For fixed h , the likelihood-ratio testing problem between the two Gaussian paths depends on

$$S_h = \sum_{r=1}^h r^{2H}$$

and has exact Bayes error $\Phi(-\beta\sqrt{S_h}/\sigma)$. The endpoint loss is of order βh^H , so the witness objective is

$$L_h(\beta) = \beta h^H \Phi(-\beta\sqrt{S_h}/\sigma).$$

In the large-ratio regime $\sigma/\zeta \rightarrow \infty$, use $S_h \sim h^{2H+1}/(2H+1)$ and set $x = \zeta\sqrt{S_h}/\sigma$. After normalizing by $\sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}$, the objective becomes $(2H+1)^{H/(2H+1)}x^{p_H}\Phi(-x)$. Differentiating gives the stated scalar equation and constant. $\square$

The ramp is therefore a valid structural witness, but it is not the best endpoint-saturating Hölder shape when $H < 1$.

Proposition 2.7 (Endpoint-minimal witness). *Fix $H \in (0, 1]$ and $h \geq 1$. Among all discrete profiles $g = (g_0, \dots, g_h)$ satisfying*

$$g_0 = 0, \quad g_h = h^H, \quad |g_r - g_s| \leq |r - s|^H,$$

the profile

$$g_r^{\min} = h^H - (h-r)^H$$

minimizes the testing energy $\sum_{r=1}^h g_r^2$. Its asymptotic energy constant is

$$I_H = \int_0^1 (1 - (1-u)^H)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

Consequently the exact Gaussian endpoint-minimal witness has constant

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H),$$

where x_H is the root from proposition 2.6. For $H < 1$, this strictly improves the ramp constant by the factor

$$\left(\frac{H+1}{2H^2}\right)^{H/(2H+1)}.$$

Proof. For any feasible profile, the endpoint constraint and Hölder condition imply

$$g_r \geq h^H - (h - r)^H.$$

The displayed $g^{\min}$ attains this lower envelope and is feasible because $x \mapsto x^H$ is subadditive on $(0, \infty)$ for $H \leq 1$. It is therefore pointwise dominated by every feasible profile and minimizes $\sum_{r=1}^h g_r^2$. The Riemann-sum limit gives

$$h^{-(2H+1)} \sum_{r=1}^h (h^H - (h - r)^H)^2 \rightarrow I_H.$$

Substituting a general energy constant I into the exact Gaussian reduction gives $I^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H)$. Taking $I = I_H$ yields the endpoint-minimal constant; comparing it with the ramp energy constant $(2H + 1)^{-1}$ gives the improvement factor. At $H = 1$, the two profiles coincide. $\square$

These refinements do not claim a class-tight minimax theorem for all deterministic Hölder paths. Their role is more precise: within the Gaussian witness program, they replace the earlier conservative testing relaxation and show that the ramp shape itself is not extremal for $H < 1$.

2.6 Beyond the Minimum Kernel

The one-dimensional minimum kernel is the first rigorous carrier instantiation. Two broader settings are also important for interpreting the law.

Extended regime. In low-dimensional or low-intrinsic-dimensional settings, the natural question is whether the triangular-array carrier inherits the same exponent as the i.i.d. mixture benchmark up to a constant-level gap. The clearest evidence comes from embedded $k = 1$ support inside a larger ambient space, where triangular and i.i.d. slopes remain near the $a = 1/2$ regime under fixed span.

Operational regime. For a practical geometry such as fixed- ε debiased Sinkhorn, the corresponding question is whether the induced finite-sample carrier is stable enough to feed the same horizon law. The experiments reported below are consistent with that interpretation: across the tested regularization values, they show a stable carrier signal rather than a qualitative change of behavior.

Together, these settings show how the abstract law extends beyond the one-dimensional minimum kernel while keeping the formal claims within their proper scope.

Corollary 2.8 (EWMA analogue). *For exponential weights $w_j = \alpha(1 - \alpha)^j$, the effective lag is $\tau_\alpha = (1 - \alpha)/\alpha$ and the effective sample size scales as $n_{\text{eff}} \asymp 1/\alpha$. Balancing the same carrier and staleness terms therefore yields the same exponent in the effective-memory scale. In particular, on the canonical regime $a = 1/2$, $H = 1$, the EWMA analogue of the cube-root horizon is recovered up to constants.*

3 Empirical Evidence

The empirical section makes the theoretical picture visible in finite samples. Its role is twofold: to show the main signatures of the variance–staleness law, and to calibrate the broader extended and operational regimes beyond the minimum-kernel result.

Four signatures matter. First, horizon sweeps should exhibit a finite optimum rather than a boundary solution. Second, that optimum should move with temporal roughness. Third, carrier inheritance should remain visible beyond the minimum kernel in extended and operational regimes. Fourth, temporal validity can fail before changepoint evidence becomes statistically visible.

3.1 A Visible Finite-Memory Optimum

The Gaussian horizon sweep isolates the variance–staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the minimum lies at an interior horizon rather than at either boundary.

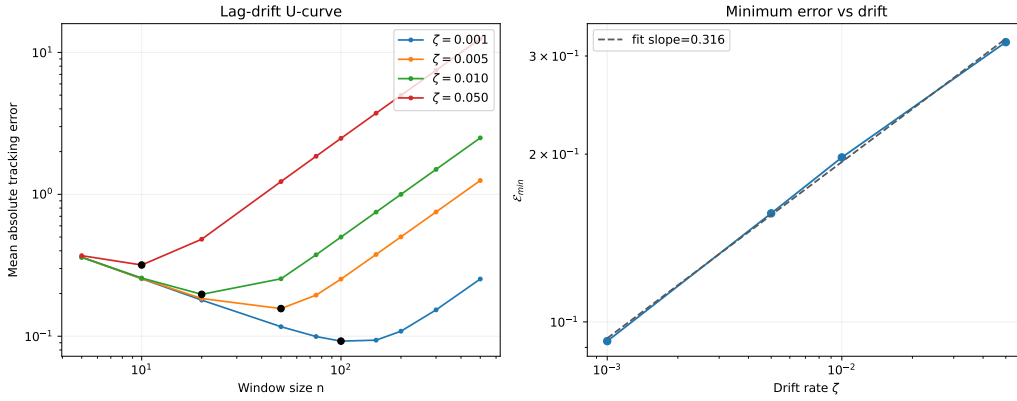


Figure 3: Structural U-curve for finite-memory tracking. Tracking error first falls with horizon and then rises once staleness overtakes the remaining variance gain. The optimum remains finite across drift strengths.

This is the most basic finite-memory signature. Under stationarity, more memory would continue to help. Under drift, the same horizon that reduces variance also ages the evidence.

3.2 Roughness-Dependent Horizon Scaling

The next experiment varies the roughness index directly. For each Hölder-type regime, the surrogate error obeys the same structural decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^H,$$

and the experiment records the empirically optimal horizon across a sweep of ζ values.

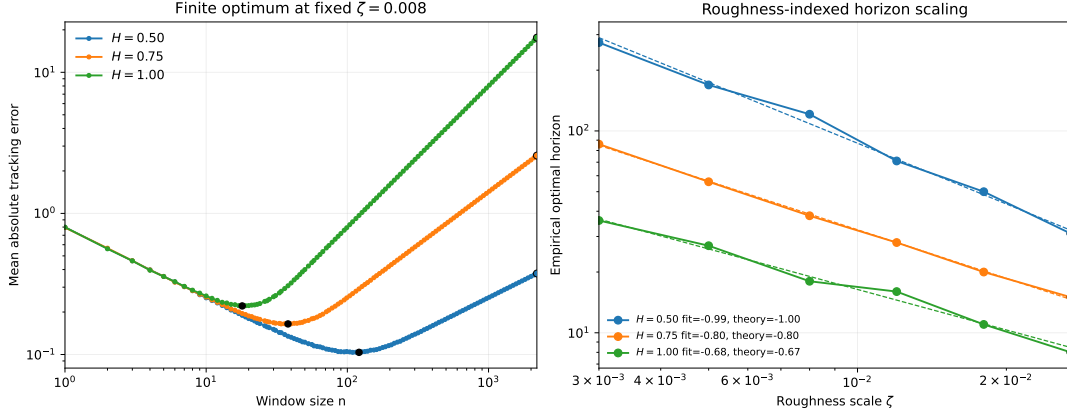


Figure 4: Roughness-dependent horizon scaling. Each path class still exhibits a finite optimum, but the location of that optimum changes with H . The worst-case Lipschitz cube-root law appears as the $H = 1$ member of a broader carrier-roughness useful-memory horizon family.

This separation is structural. If the cube-root law were the only story, the right panel would collapse to a single slope. It does not. The empirical optima separate by roughness class, matching the path-dependent horizon logic of corollary 2.2.

3.3 Extended and Operational Carrier Regimes

The next figure summarizes two broader carrier settings: an extended regime based on low intrinsic dimension, and an operational regime based on fixed- ε Sinkhorn geometry.

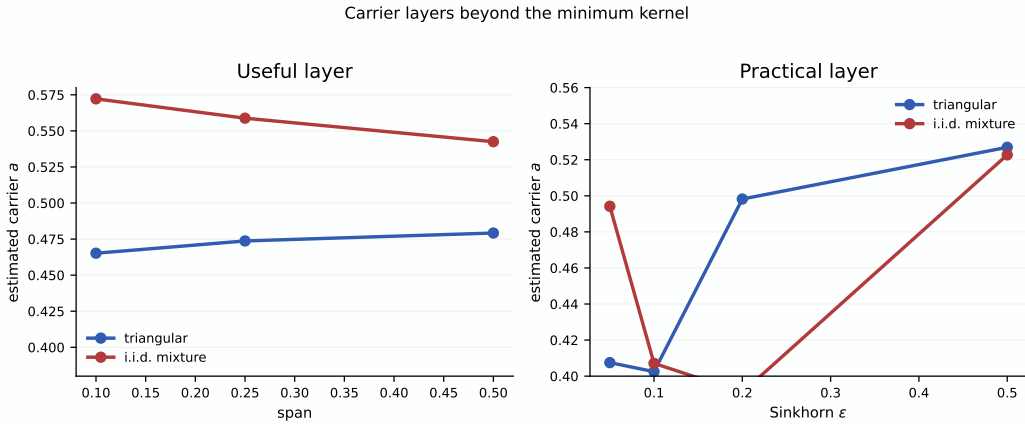


Figure 5: Extended and operational carrier settings. Left: in the extended regime, embedded low-intrinsic-dimensional support preserves a carrier near the $a = 1/2$ regime, and triangular slopes stay close to the i.i.d. mixture benchmark. Right: in the operational regime, fixed- ε Sinkhorn produces a carrier whose triangular and i.i.d. slopes remain qualitatively close across the tested ε values.

The left panel concerns bounded-support fixed-span samples with low intrinsic

dimension. In that setting, the triangular-array carrier tracks the i.i.d. mixture benchmark rather than exhibiting a visibly different exponent.

The right panel concerns fixed- ε Sinkhorn geometry. It shows that the corresponding carrier remains stable across the tested regularization values and can therefore be read against the same abstract horizon law. Appendix A reports the corresponding summary table.

3.4 Detector-Silent Staleness

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over time. The useful-memory horizon n_t^* contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\text{inv}} := t_{\text{detect}} - t_{\text{valid}},$$

where t_{valid} is the first time the operating horizon persistently exceeds the useful-memory scale, and t_{detect} is the first detector alarm that follows.

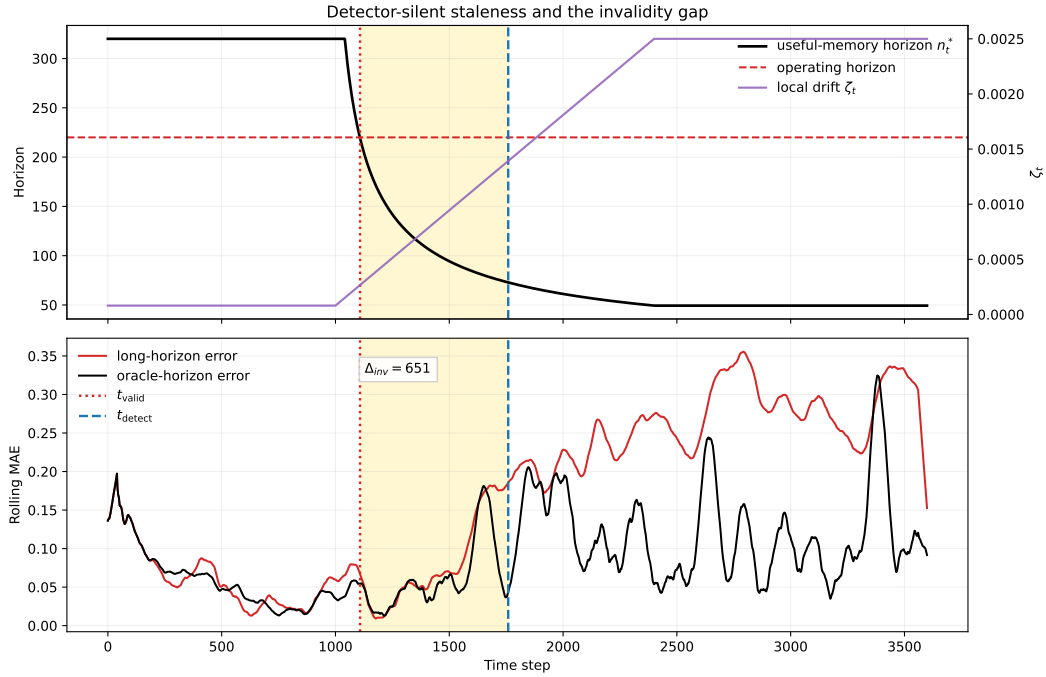


Figure 6: Detector-silent staleness. As drift intensifies, the useful-memory horizon contracts until the operating horizon becomes stale at t_{valid} ; the detector responds only later at t_{detect} , leaving a strictly positive invalidity gap. The lower panel shows excess tracking error accumulating before the alarm appears.

Across the reported runs, the measured gap remains positive. This is the empirical form of detector-silent staleness: retained evidence can already be invalid for the present while detector statistics remain below alarm threshold.

Table 1: Detector-threshold ablation for the invalidity gap. Across the tested ADWIN thresholds, mean detection time remains strictly later than mean validity-expiry time, so the measured invalidity gap stays positive throughout.

δ	$\bar{t}_{\text{valid}}$	$\bar{t}_{\text{detect}}$	$\bar{\Delta}_{\text{inv}}$
0.0005	1108.0	1671.0	563.0
0.0010	1108.0	1657.7	549.7
0.0020	1108.0	1641.7	533.7
0.0040	1108.0	1631.0	523.0

Table 2: Detector-family comparison for the invalidity gap. Using the same drift ramp and operating horizon, the positive gap persists for both ADWIN and Page–Hinkley, even though their changepoint clocks differ.

Detector	$\bar{t}_{\text{valid}}$	$\bar{t}_{\text{detect}}$	$\bar{\Delta}_{\text{inv}}$
ADWIN	1108.0	1641.7	533.7
Page–Hinkley	1108.0	1410.6	302.6

3.5 Horizon Misalignment Cost

The final empirical question is how sharply performance degrades when the operating horizon misses the useful-memory scale. Using the same roughness-indexed surrogate, fig. 7 plots excess error against the relative horizon gap $|\log(n/n^*)|$.

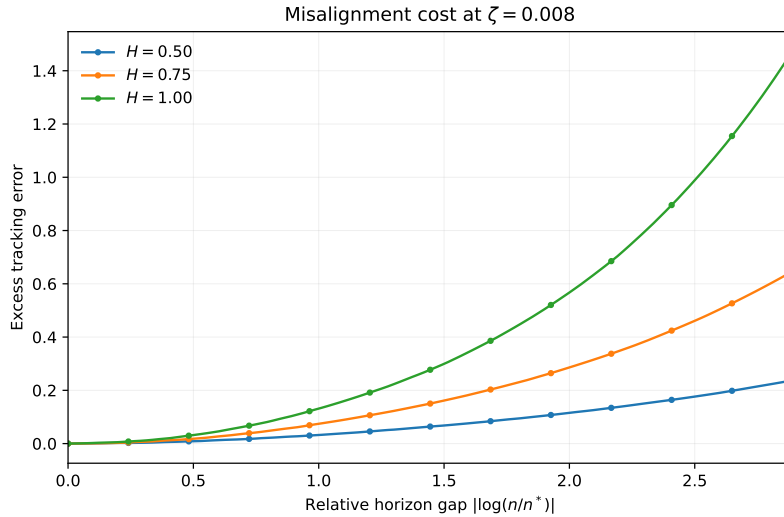


Figure 7: Cost of horizon misalignment. Excess tracking error is smallest near the useful-memory scale and rises once the operating horizon moves away from it. Being too short is noisy; being too long is stale.

This is the operational reason the horizon matters. The optimum is not only finite; it is a scale around which performance is visibly better than on either side.

3.6 What the Evidence Shows

Taken together, the experiments support four claims that match the theory. Finite-memory tracking exhibits a structural U-curve rather than an ever-growing memory benefit. The useful-memory scale is roughness dependent, with the cube-root law appearing as the $H = 1$ member of a broader family. Carrier stability remains visible beyond the minimum kernel in both extended and operational settings. Temporal validity can also fail before changepoint evidence appears, producing a measurable invalidity gap.

4 Related Work

Nonstationary optimization and dynamic regret. Dynamic-regret and variation-budget results show that under drift, window size or forgetting scale must depend on how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. Those works are close in spirit, but their primary object is regret under a moving comparator. Here the primary object is different: the temporal validity of retained evidence in a distribution-tracking problem. The horizon is not a tuning parameter attached to regret analysis; it is the statistical object being characterized.

Adaptive windowing and drift detection. Adaptive-window and concept-drift methods ask when accumulated evidence is sufficient to react [Bifet and Gavalda, 2007, Gama et al., 2004, 2014]. Hinder et al. [2023] emphasize that the relevant window scale is itself part of the problem. The present question is adjacent but distinct. Temporal validity is not changepoint evidence: memory can stop representing the present before the available data support a detector alarm. In that sense, the paper is about the validity horizon of retained evidence rather than the stopping time of a reaction rule. The invalidity gap makes that distinction visible in finite samples.

Adaptive filtering and state-space tracking. Classical adaptive filtering and state-space tracking also balance smoothing against responsiveness by tuning forgetting factors, gain schedules, or process noise models [Kalman, 1960, Jazwinski, 1970]. Those analyses usually work under stochastic state-space assumptions and ask for Bayes-optimal tracking of a latent process. The present paper instead studies deterministic Wasserstein drift envelopes and characterizes a worst-case temporal-validity horizon.

Locally stationary smoothing. Locally stationary time-series analysis likewise studies how window or bandwidth choice must adapt to time variation [Dahlhaus, 1997]. That literature is close in its bias-variance logic, but the moving object there is typically a local spectrum or regression function. Here the moving object is a path of distributions, and the age penalty is measured directly in transport geometry.

Distribution tracking under transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural when drift lives on paths of distributions rather than on scalar losses [Fournier and Guillin, 2015, Genevay et al., 2019, Niles-Weed

and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. They supply the geometric language in which staleness becomes a transport error between past and present. In this setting, the trade-off is not simply between adaptation and variance, but between finite-sample accuracy and temporal misalignment measured in distributional geometry.

Wasserstein nonstationarity and robust optimization. Jiang, Li, and Zhang study online stochastic optimization under a Wasserstein nonstationarity measure and obtain distribution-aware regret guarantees [Jiang et al., 2020]. Keehan, Anderson, and Wiesemann analyze related drift-aware weighting questions in Wasserstein distributionally robust optimization [Keehan et al., 2025]. Amini and Vinas give a recent moment-method route to triangular-array concentration in W_1 [Amini and Vinas, 2026]. Those results are adjacent to the carrier-regime issue studied here, but they do not by themselves close the present W_2 window-mixture concentration problem. The present focus is narrower and more structural: for finite-memory tracking, how long can retained evidence continue to improve estimation of the current distribution before age becomes a dominant error source?

Freshness and Age of Information. The Age-of-Information literature treats age as a resource constraint and asks how update policies balance communication cost against stale information [Kaul et al., 2012, Yates et al., 2021]. The present paper gives a distributional analogue. Age is costly because it induces geometric error under drift, and temporal roughness determines how quickly that cost accumulates.

Across these neighboring literatures, the closest theme is that nonstationarity makes retention scale matter. The distinguishing claim here is more specific: finite memory obeys a carrier-roughness useful-memory horizon law, the worst-case cube-root law is the $a = 1/2$, $H = 1$ member of that family, and temporal validity can fail before change becomes statistically detectable. The resulting viewpoint is complementary to regret and detection analyses: it characterizes when memory itself ceases to be a statistically valid proxy for the present.

5 Limitations and Scope

The central results of the paper are the carrier-roughness useful-memory horizon law, the minimum-kernel carrier on the canonical $a = 1/2$ regime, and the structural lower bound on that same regime. The refined Gaussian witness results sharpen the constants inside that regime, but broader extensions are stated more cautiously.

Carrier scope. The theory is modular in the carrier exponent a , but the main rigorous regime is the root- n carrier. The minimum kernel establishes that regime in one dimension under bounded support and fixed span. Outside that setting, the same balancing logic continues to hold, but the carrier must come from a different theorem. In particular, the paper does not claim a general raw- W_2 triangular-array carrier theorem in arbitrary dimension.

Extended and operational regimes. The extended setting and the fixed- ϵ Sinkhorn setting are supported here primarily through finite-sample evidence. The

low-intrinsic-dimensional results indicate that the triangular-array carrier can track the i.i.d. mixture benchmark, and the Sinkhorn experiments indicate a stable carrier signature across the tested regularization values. A theorem of the same strength as the minimum-kernel result is not established here for either setting.

Lower-bound scope. The lower bound is subclass based. The Gaussian witness is enough to show that on the canonical carrier regime $a = 1/2$, the useful-memory scale is structural rather than an artifact of a particular estimator. What remains open is class-tightness and constant-sharpness for full deterministic Hölder path classes, especially away from the canonical regime and beyond the present witness geometry. The current witness frontier is already sharper than the original ramp-plus-Pinsker sketch: the exact Gaussian testing bound improves the canonical-regime constant, and for $H < 1$ the endpoint-minimal Hölder witness is stronger than the ramp witness. But those are still subclass refinements, not class-tight minimax theorems.

Weights and memory shape. The main text emphasizes uniform windows, with an EWMA corollary showing that the same exponent logic survives after effective-lag reparameterization. The paper does not investigate whether more general non-uniform weights can improve constants or alter exponents across broader path classes.

Empirical scope. The experiments are synthetic and structurally motivated. Their role is to make the theory visible and to calibrate broader carrier settings, not to demonstrate competitive performance on a benchmark suite. Real-data validation and direct comparison with adaptive horizon-selection rules remain outside the present scope.

Detectability versus validity. Detector-silent staleness is an empirical consequence of the structural theory, not the primary theorem object. The paper shows that temporal validity can fail before detector alarms appear, but it does not prove a universal detection-delay theorem across detector classes.

Open problems. The next technical questions are clear. The first is to establish carrier inheritance beyond the minimum kernel under natural low-dimensional conditions. The second is to understand when practical geometries, such as fixed- ε Sinkhorn, admit theorem-level carrier bounds. The third is to extend the lower bound beyond the canonical $a = 1/2$ regime and decide how far the new Gaussian-witness refinements can be pushed toward class-tight lower theory. After that come non-uniform weighting, real-data validation, and stronger stochastic-process formulations.

6 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting trade-off makes useful memory a finite-horizon object rather than an unlimited statistical resource.

The paper formalizes that idea through a carrier-roughness useful-memory horizon law. The central theorem is the abstract decomposition

$$\mathbb{E} d\left(\widehat{P}_t^{(n)}, P_t\right) \leq C_K n^{-a} + C_S \zeta n^H,$$

whose optimized scale is

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The cube-root horizon is therefore one member of a broader law: the special case $a = 1/2$, $H = 1$ corresponding to a root- n carrier under worst-case linear staleness.

The analysis anchors this law in a rigorous minimum-kernel result and a structural lower bound on the same canonical carrier regime. Together they show that, in the bounded-support fixed-span one-dimensional setting, the horizon is neither an artifact of a particular estimator nor a purely heuristic balance. The refined Gaussian witness analysis sharpens that conclusion: exact testing gives a clean scalar frontier, and the endpoint-minimal Hölder witness improves the ramp constant for $H < 1$. The broader extended and operational regimes show how the same law remains relevant beyond the initial proof kernel.

The empirical section makes the theory visible. It shows a structural U-curve, roughness-dependent horizon scaling, carrier behavior beyond the minimum kernel, and a strictly positive invalidity gap between temporal validity and changepoint evidence. The resulting message is both theoretical and operational: useful memory has a path-dependent horizon, and retained evidence can become invalid before change becomes statistically detectable.

The resulting shift in viewpoint is that memory under drift is not treated primarily as a regret-tuning parameter or an alarm-design parameter, but as a temporal-validity object in distributional geometry.

The main open questions concern carrier inheritance beyond the minimum kernel, theorem-level operational-geometry results, and broader lower-bound extensions beyond the canonical $a = 1/2$ regime. These are natural extensions of the present theory, not changes to its central object.

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A Finite-Sample and Carrier-Regime Diagnostics

This appendix collects the numerical diagnostics that support the carrier side of the theory. These checks are not substitutes for the theorem statements. Their role is to show that the minimum-kernel, extended-regime, and operational-regime interpretations are consistent with the finite-sample behavior of the current experiment families.

A.1 Finite-Sample Geometry Check

For the finite-sample term, we estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log–log slope in n . We also probe the non-identically distributed fixed-span window directly in one dimension by comparing a triangular-array sample, with one draw from each drifted component, against an i.i.d. sample from the same window mixture. In that check the within-window span ζn^H is held fixed, so the experiment targets the minimum-kernel carrier interpretation rather than a large-span regime in which heterogeneity itself can become dominant. Finally, we check the Sinkhorn proxy at a few fixed values of ε .

Table 3: Empirical scaling check for the finite-sample term. The bounded-support i.i.d. W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, the corresponding triangular-array check compares one sample from each drifted component against an i.i.d. sample from the window mixture, and the Sinkhorn proxy is reported at fixed ε . These rows are numerical support for the carrier-regime discussion rather than theorem statements.

Setting	Estimated slope	Comment
$W_2, d = 1$	-0.48	close to $-1/2$
$W_2, d = 2$	-0.41	slower decay
$W_2, d = 4$	-0.25	slower decay
$W_2, d = 5$	-0.24	slower decay
W_2 , i.i.d. mixture, $d = 1, H = 0.5$	-0.50	fixed span $\zeta n^H = 0.25$
W_2 , triangular, $d = 1, H = 0.5$	-0.45	fixed span $\zeta n^H = 0.25$
W_2 , i.i.d. mixture, $d = 1, H = 1.0$	-0.47	fixed span $\zeta n^H = 0.25$
W_2 , triangular, $d = 1, H = 1.0$	-0.41	fixed span $\zeta n^H = 0.25$
Sinkhorn $\varepsilon = 0.50$	-0.21	proxy constant changes
Sinkhorn $\varepsilon = 0.10$	-0.53	proxy constant changes
Sinkhorn $\varepsilon = 0.05$	-0.67	proxy constant changes

This table supports the carrier interpretation used in the main text. In low dimension, the raw i.i.d. Wasserstein term is compatible with a root- n carrier, whereas higher-dimensional raw Wasserstein slows down. In the one-dimensional fixed-span triangular-array check, the recovered slopes stay near the corresponding i.i.d. mixture baseline, which is consistent with the minimum-kernel and design-effect reading used in proposition 2.4.

A.2 Extended and Operational Carrier Regimes

The next table summarizes the broader regimes beyond the minimum kernel. The extended rows compare triangular and i.i.d. mixture carriers on embedded low-intrinsic-dimensional support across span. The operational rows compare fixed- ε Sinkhorn carriers across regularization values.

These rows should be read as calibration rather than proof. The extended-regime rows support the claim that low intrinsic dimension preserves a carrier near the canonical $a = 1/2$ regime without a dramatic triangular-versus-benchmark gap.

Table 4: Summary rows for the extended and operational carrier regimes. The extended rows compare triangular and i.i.d. mixture carriers across span, while the operational rows compare fixed- ϵ Sinkhorn across ϵ .

Regime	Setting	Parameter	Carrier a
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.10	0.10	0.4652
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.10	0.10	0.5722
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.25	0.25	0.4737
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.25	0.25	0.5588
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.50	0.50	0.4792
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.50	0.50	0.5425
operational	Sinkhorn iid mixture $\epsilon=0.50$	0.50	0.5227
operational	Sinkhorn triangular $\epsilon=0.50$	0.50	0.5269
operational	Sinkhorn iid mixture $\epsilon=0.20$	0.20	0.3910
operational	Sinkhorn triangular $\epsilon=0.20$	0.20	0.4982
operational	Sinkhorn iid mixture $\epsilon=0.10$	0.10	0.4071
operational	Sinkhorn triangular $\epsilon=0.10$	0.10	0.4025
operational	Sinkhorn iid mixture $\epsilon=0.05$	0.05	0.4942
operational	Sinkhorn triangular $\epsilon=0.05$	0.05	0.4075

The operational-regime rows support the claim that fixed- ϵ Sinkhorn acts as a stable carrier in the current synthetic setting, even though the paper does not claim a general theorem for that regime.

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